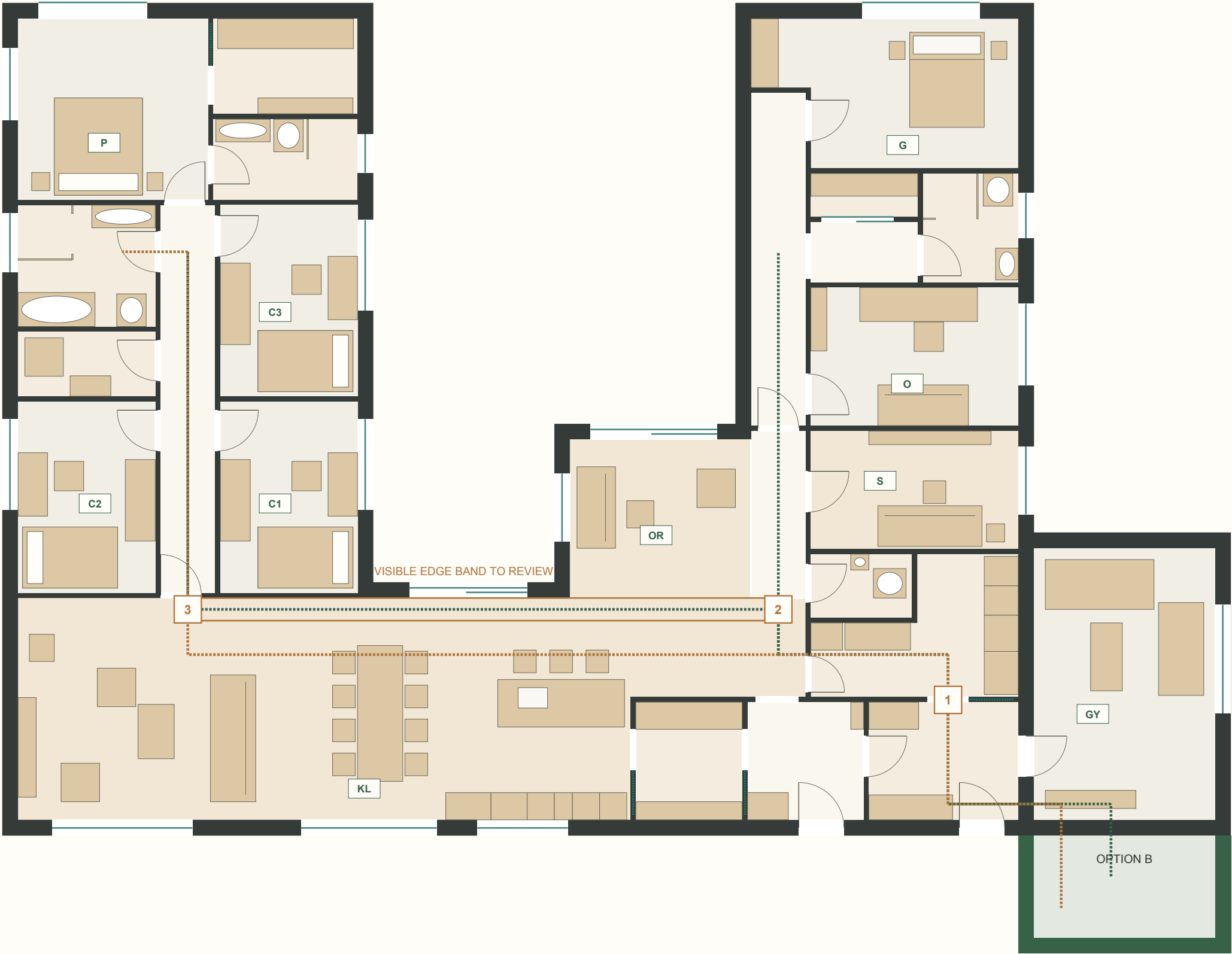


The difficult connection is across the shared-room vault

Route study 1:100 at A3 | Uses unselected plant option B for comparison | South up / north down



1 / PLANT TO SERVICE WING

Trace above the gym / service rooms.
Keep access to filters and connections.
Check crossing of the retained gym wall.
Do not use the gym floor for equipment.

2 / GUEST WING BRANCH

Guest hall offers a candidate route.
Check duct + silencer dimensions.
Coordinate office and snug sound.
No terminal positions are selected.

3 / FAMILY WING CROSSING

No existing enclosed corridor spans
the open living / dining room.
A visible service band may be needed.
Section on page 2 exposes the effect.
Concealment is not yet resolved.

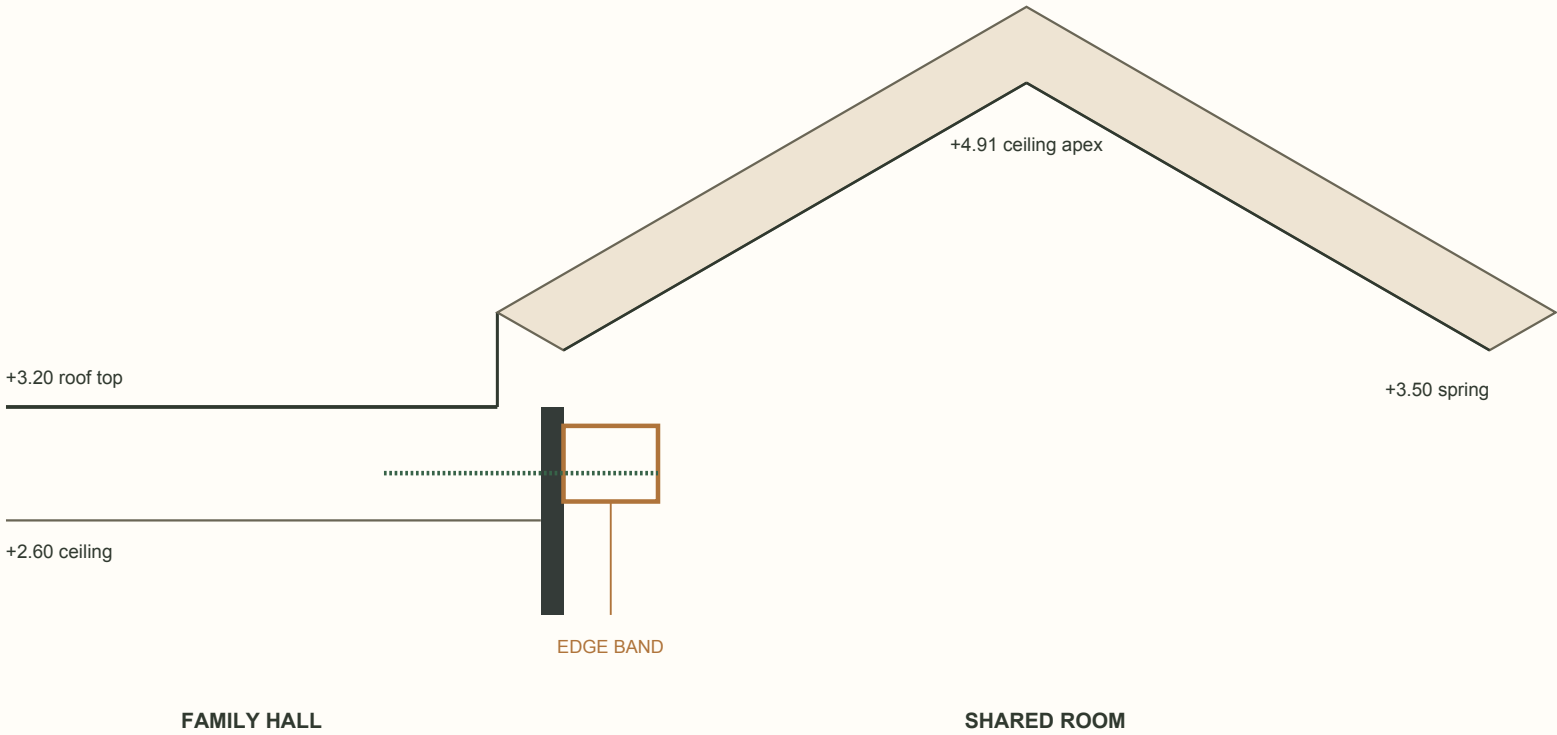
AMBER / HOT WATER

35.2 m drawn plan route to family basin.
No verticals, fittings or final branches.
Compare waiting time and heat loss;
size pipes before choosing circulation.

Green = conceptual ventilation / cooling distribution. Amber = conceptual water route and unresolved crossing. Lines are not installed duct sizes.

Keep the vault honest about the space services need

Longitudinal section through the family-hall / shared-room junction at x = 4.10 m | 1:40 at A3



A RESERVATION, NOT A HIDDEN SOLUTION

Test a 0.50 m projection x 0.40 m deep edge band.

Its underside is +2.70 m; top +3.10 m.

It would be visible below the shared-room vault.

The lighting ledge and room proportions need review.

No enclosure is added to the current plan.

LOWER ROOF / CEILING ZONE

Roof top +3.20 m minus ceiling +2.60 m = 0.60 m gross.

Roof structure, insulation, falls and ceiling consume this.

Do not count 0.60 m as a usable duct void.

Header penetrations and door head remain unresolved.

No structural member may be assumed drillable.

COMPARE BEFORE CHOOSING

A: accept an accessible service band at the shared edge.

B: investigate distributed ventilation / short cooling runs.

B trades a major crossing for more local equipment.

Both need capacities, noise, terminals and access checked.

Neither distribution strategy is selected.

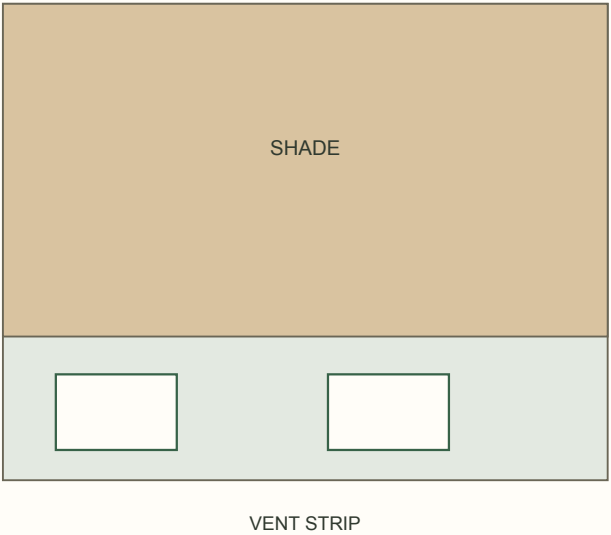
SECTION STATUS

Roof heights follow the existing concept 06B direction. The header and services band are coordination reservations only.

This section identifies a design clash; it does not resolve roof construction, duct sizing or architectural acceptance.

Coordinate the plant height, roof drainage and glazed roof together

Plant wall elevation 1:25 at A3 | Roof / shade diagram 1:50 at A3 | Provisional dimensions



GLAZED ROOF / RETAINED CONCEPT 11

Roof shade stops before the high vent reservations.

Approximately 0.95 m of upper glass remains unshaded.

Plan rectangles do not overlap; hardware may still conflict.

Check opening arcs, cassette, runoff and maintenance.

Include the unshaded strip in overheating calculations.

PLANT AND GYM ROOF

Continue the lower-block roof direction, with falls and outlets designed for the enlarged catchment.

Check foundations, equipment weights, vibration, moisture and service / replacement access.

Select outdoor-unit locations, air intake / exhaust separation and condensate / discharge routes against the actual site.

PV, battery and inverter locations are open; do not count them as fitted in this plant room.

Design comfort for each room, then test the solar contribution

Required room coverage and calculations | No capacities inferred from floor area alone

SPACE	CLEAR AREA	COORDINATION PRIORITY
Parents bedroom	16.80 m2	Quiet night cooling; avoid bed draughts.
Child 1	12.79 m2	Night comfort, blackout and desk / bed air paths.
Child 2	12.79 m2	Night comfort, blackout and desk / bed air paths.
Child 3	12.79 m2	Night comfort, blackout and desk / bed air paths.
Guest bedroom	17.12 m2	Guest night comfort and local controls.
Kitchen / dining / living	74.73 m2	Vaulted volume, cooking gains and glazing.
Open garden sitting area	14.00 m2	Connected to KL; roof glass and shade dominate.
Office / gaming	13.97 m2	Computer gains; work and overnight guest modes.
Snug	11.91 m2	Quiet film use; avoid screen reflections / fan noise.
Indoor gym	24.00 m2	Exercise gains, fresh air and clear equipment zones.

HEATING / HOT WATER / VENTILATION

Room heat loss follows confirmed fabric, weather and temperatures.
Hot water: two simultaneous high-flow showers, then repeat-use recovery.
Design extract in kitchen, bathrooms, WC and laundry; balance supply.
Keep MVHR duties separate from required active cooling capacity.

SOLAR / BATTERY / GRID

Match hourly PV generation to cooling and other household loads.
Include evening / night cooling and the office guest mode.
Compare direct PV use, storage losses, battery power and capacity.
Report grid imports and solar coverage; no solar-only claim yet.

A concrete brief for the next design stage

Inputs, owners and outputs | Prepare these before equipment or joinery is ordered

SITE / ARCHITECT	Survey, levels, boundaries, orientation, exposure and access.
	Locate the extension, outdoor units, carport / PV and drainage.
ARCHITECT / STRUCTURE	Wall, floor, roof, glazing and airtightness assumptions.
	Resolve roof junctions, the service crossing and maintenance access.
HEATING / WATER DESIGNER	Temperatures, fabric, shower flows / duration, incoming supply.
	Room heat loss, hot-water storage / recovery and distribution design.
VENTILATION / COOLING DESIGNER	Occupancy, room gains, acoustic targets and control preferences.
	Airflows, cooling loads, terminals, silencers and condensate routes.
SOLAR / ELECTRICAL DESIGNER	Array areas, shading, load profile and battery / EV preferences.
	Hourly energy balance, supply capacity and protected equipment siting.
COST REVIEW	Chosen scope and coordinated quantities.
	Compare capital / running cost and maintenance before selection.

CURRENT PRIMARY REFERENCES / CHECKED 13 SEPTEMBER 2026

Energy Saving Trust: room-by-room heat loss

Zehnder: ventilation system selection and duties

Energy Saving Trust: battery storage

GOV.UK: Approved Document O; applicability / edition to confirm for project